

## **Limitations of JDBC (Java Database Connectivity)**

### **1. SQL Query Language Dependency:**

- Developers must manually write and manage SQL queries for all database operations.
- This increases the complexity and potential for errors, especially in large applications with complex queries.

### **2. Repetitive Code:**

- JDBC requires boilerplate code for tasks such as establishing connections, creating statements, and handling exceptions.
- The repetitive nature of this code can lead to increased development time and maintenance challenges.

### **3. Lack of Named Parameters:**

- JDBC only supports positional parameters in prepared statements (e.g., ? placeholders), making it harder to manage queries with multiple parameters.
- Named parameters, which allow parameters to be identified by name rather than position, are not supported, reducing the readability and flexibility of queries.

These limitations often lead developers to seek out higher-level frameworks like Hibernate, which abstract away much of the complexity involved in database interactions and provide more features such as named parameters and automatic SQL generation.